

# What a Filing Reveals: A Companion Model for the Electoral Costs of Judicialization

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## Abstract

Adversarial electoral litigation widens the top-two margin in Brazilian mayoral races. This note asks what mechanism can produce that, and finds that the natural candidate cannot. In the transfer class of campaign-attack models, an attack moves support from its target to its sender; under the attack direction those models derive and document, the map is a contraction on the top-two margin, so it cannot generate the result. The alternative developed here turns on what a filing reveals. A filing is a noisy signal about a candidate's type and a precise signal about a candidate's rank, because the filer is optimizing and targets the leading opponent. Litigation therefore informs voters about position more than about quality. Movement in a single primitive, the filing rate, then delivers three comparative statics – resting on two sign restrictions on how that rate moves the arena's two precisions, and, for the third, on prior precision about the leader differing by seat type. The posterior gap between the top two becomes less dispersed, so rank reversals become rarer and the expected margin widens. The posterior variance of every candidate's valence rises, so an ambiguity-averse voter is readier to leave the valid ballot, which under compulsory voting means a blank or null vote. And because an unsettled record disperses voters' impressions, the runner-up's supporters are selected on idiosyncratic distaste for the leader only where the leader is an incumbent – which is what separates open from contested seats.

## 1 Environment

One mayoral race. Candidates  $j = 1, \dots, J$  are indexed by pre-shock support  $s_1^\circ \geq s_2^\circ \geq \dots \geq s_J^\circ$ , and there is a continuum of voters of mass one. Voter  $i$ 's payoff from candidate  $j$  is

$$u_{ij} = \hat{\theta}_j - \frac{\rho}{2} \hat{V}_i(\theta_j) + \varepsilon_{ij},$$

where  $\theta_j$  is a common-value valence nobody observes and  $\varepsilon_{ij} \sim N(0, v_\varepsilon)$ , drawn independently across voters and candidates, is a durable idiosyncratic taste the voter knows.  $\hat{V}_i(\theta_j)$  is the posterior variance of  $\theta_j$ ; every voter holds the same one, so we write  $\hat{V}(\theta_j)$ . The coefficient  $\rho \geq 0$  prices it: the term is the mean–variance reduction of a weight-of-evidence preference, under which thin evidence about a candidate is itself a reason to withhold a choice (Ghirardato and Katz, 2006). At  $\rho = 0$  the voter is an expected-utility maximizer and every statement below that runs through  $\rho$  vanishes, which is what makes the ambiguity content testable rather than assumed.

Beliefs about valence start at  $\theta_j \sim N(m_j, 1/h_j)$ :  $m_j$  is the candidate's reputation and  $h_j$  measures how settled the record is. An incumbent who has governed for four years has a high  $h$ ; a first-time candidate has a low one.

How settled a record is also governs how much voters agree about it.

**Assumption 1** (Impressions). Before the campaign, voter  $i$ 's impression of candidate  $j$  is  $m_j + \zeta_{ij}$  with  $\zeta_{ij} \sim N(0, \kappa_\zeta/h_j)$ , independent across voters and candidates and independent of  $\varepsilon$ . Impressions decide which candidate a voter starts out supporting. They do not survive the campaign: a filing wave is public and draws local attention, so by the ballot stage the common posterior  $\hat{\theta}_j$  has displaced them, and the voter's ballot-stage payoff carries  $\varepsilon_{ij}$  but not  $\zeta_{ij}$ .

A four-year record leaves little room for disagreement about what a mayor is like; a fresh candidate leaves a great deal. That is the content of  $\kappa_\zeta/h_j$ . The second half of the assumption is the half that does work: it separates what sorts voters into camps before the campaign (impression plus taste) from what they act on at the ballot (taste alone). It is not free, and Section 6 says what it costs.

Let  $\bar{b}$  denote the payoff from a blank or null vote, which is what refusing to endorse anyone is worth once the ballot carries no instrumental value.

Treatment is  $\lambda$ , the intensity of adversarial filings in the race, and it is the object the shift-share instrument shifts. Two institutional facts fix the timing and the margin of adjustment. Candidate registration closes on 15 August while adversarial filings run through October, so the candidate set is predetermined with respect to  $\lambda$ . And voting is compulsory (CF/88 art. 14 §1), so a voter who wants out of the contest adjusts the mark rather than the attendance: the exit option is a blank or null vote.

Throughout,  $\varphi$  and  $\Phi$  are the standard normal density and distribution function, and  $\bar{\Phi} = 1 - \Phi$ . The attack fraction  $\phi$  of Section 2 is a separate object and does not reappear after it. Precisions are written  $h$  and  $\tau$ , variances  $v$  and  $\hat{V}$ , and standard deviations  $\sigma$  and  $\varsigma$ : so  $\bar{\tau}$  and  $\tau_{\text{status}}$  below are precisions, while  $\sigma_\Delta$  and  $\sigma_D$  are standard deviations.

## 2 The transfer channel cannot produce the result

### 2.1 The map

Nakaguma and Souza (2025) model a campaign attack as a transfer. An attack from  $k$  on  $j$  moves a fraction  $\phi$  of  $j$ 's support away from  $j$ ; in their Appendix G.2 demobilization extension a fraction  $\alpha$  of that mass leaves for the outside option instead of reaching  $k$ , and the attacker loses a fraction  $\gamma$  of its own support to backlash.

Their equilibrium has every candidate target the highest-ranked opponent, so in a three-candidate race the profile is  $1 \rightarrow 2$ ,  $2 \rightarrow 1$  and  $3 \rightarrow 1$ : mutual between the top two, with the third attacking the leader. Their data agrees. Measuring attacks as electoral lawsuits, the observed frequencies in three-candidate races are  $2 \rightarrow 1$  at 3.0pp,  $1 \rightarrow 2$  at 1.9pp and  $3 \rightarrow 1$  at 0.8pp. The second-place candidate is the most aggressive and the front-runner is the most attacked.

### 2.2 The contraction

**Proposition 1** (Transfer cannot consolidate). *Take  $\phi > 0$ ,  $\alpha \leq 1$ ,  $\gamma \geq 0$  and the feasibility constraint  $\phi + \gamma \leq 1$ , which says a candidate cannot lose more support than it has. Under mutual  $1 \leftrightarrow 2$  attacks, post-attack support is*

$$s_1 = s_1^\circ(1 - \phi - \gamma) + \phi(1 - \alpha)s_2^\circ, \quad s_2 = s_2^\circ(1 - \phi - \gamma) + \phi(1 - \alpha)s_1^\circ,$$

*so the top-two margin satisfies*

$$s_1 - s_2 = (s_1^\circ - s_2^\circ)\kappa, \quad \kappa \equiv 1 - \gamma - \phi(2 - \alpha) < 1.$$

Adding  $3 \rightarrow 1$  subtracts a further  $\phi s_1^\circ$  from the margin. Hence the transfer map cannot widen the top-two margin while preserving the identity of the leader. The conclusion survives normalizing by the surviving valid vote, and it survives letting the two attack intensities differ.

*Proof. Step 1 (the contraction).* Subtracting,

$$s_1 - s_2 = (s_1^\circ - s_2^\circ)(1 - \phi - \gamma) - \phi(1 - \alpha)(s_1^\circ - s_2^\circ) = (s_1^\circ - s_2^\circ)[1 - \gamma - \phi(2 - \alpha)].$$

Since  $\alpha \leq 1$  we have  $2 - \alpha \geq 1$ , so  $\phi(2 - \alpha) \geq \phi > 0$ , and with  $\gamma \geq 0$  this gives  $\kappa < 1$ . If  $\kappa \in (0, 1)$  the margin cannot widen: it narrows strictly when  $s_1^\circ > s_2^\circ$ , and it stays at zero when  $s_1^\circ = s_2^\circ$ , which the weak ordering of Section 1 admits. If  $\kappa \leq 0$  the sign of  $s_1 - s_2$  is non-positive, so candidate 2 has caught or passed candidate 1 and the leader is no longer the pre-attack leader. When  $3 \rightarrow 1$  is added, candidate 1 loses a further  $\phi s_1^\circ$  and candidate 2 loses nothing, so  $s_1 - s_2 = (s_1^\circ - s_2^\circ)\kappa - \phi s_1^\circ$ , which is smaller still.

*Step 2 (shares, not levels).* The map leaks mass: a fraction  $\alpha$  of every transfer goes to the outside option and a fraction  $\gamma$  of each attacker's own support is lost to backlash. So  $s_1 - s_2$  is a level margin while the estimates are vote-share margins, and the claim has to be checked after normalization. Write  $S = s_1^\circ + s_2^\circ$  and leave  $s_3^\circ$  untouched. Then  $s_1 + s_2 = S(1 - \gamma - \phi\alpha)$  and the surviving total is  $T = S(1 - \gamma - \phi\alpha) + s_3^\circ$ . The normalized margin widens only if  $\kappa/T > 1/(S + s_3^\circ)$ , that is only if

$$2\phi(\alpha - 1)S > s_3^\circ(1 - \kappa).$$

The left side is at most zero because  $\alpha \leq 1$ ; the right side is at least zero because  $1 - \kappa = \gamma + \phi(2 - \alpha) > 0$ . The inequality fails, so the normalized margin cannot widen either.

*Step 3 (asymmetric intensities).* Let  $\phi_{12}$  and  $\phi_{21}$  be the fractions the leader and the runner-up strip from each other. Then

$$s_1 - s_2 = (s_1^\circ - s_2^\circ)(1 - \gamma) - (2 - \alpha)(\phi_{21}s_1^\circ - \phi_{12}s_2^\circ),$$

so widening requires  $\phi_{12}/\phi_{21} > s_1^\circ/s_2^\circ > 1$ : the leader would have to attack harder than the runner-up, by more than its own support advantage. The documented intensities run the other way,  $\phi_{12}/\phi_{21} \approx 1.9/3.0$ , so the asymmetric case holds a fortiori and the closing claim covers the whole class.  $\square$

A grid check of the closed form and of the no-widening claim, including the  $3 \rightarrow 1$  profile, is in `docs/model/check_proposition1.py`.

*Remark.* Widening the margin under the transfer map requires a one-sided downward attack,  $1 \rightarrow 2$  with no reciprocation. That is excluded by the equilibrium, in which candidate 2's best target is candidate 1, and it is not what the data shows. The magnitudes do not reach the result either. An attack frequency of 3pp means the transfer occurs with probability 0.03, and conditional on occurring it moves  $\phi s^\circ$ . Even at a generous  $\phi = 0.3$  against a target holding a third of the vote, the expected move is  $0.03 \times 0.3 \times 0.33 \approx 0.003$ : about three tenths of a point, an order of magnitude below the three-point vote move being explained.

*Remark (Object of comparison).* Nakaguma and Souza (2025) work with a win probability through a logit contest function, whereas the estimates here are vote-share margins. The two are monotone in  $s_1 - s_2$ , so Proposition 1 transfers, but the mapping is an assumption and not an identity.

*Remark* (What Proposition 1 does not do). The top-two margin is an order statistic, so any mean-preserving increase in the dispersion of vote shares widens it mechanically. Proposition 1 rules out the transfer channel under the documented attack direction; it does not rule out a generic dispersion shock. What rules that out is the ballot response and the contrast between seat types in Section 5: a dispersion shock predicts neither, and predicts no directional flow from the runner-up to the leader.

### 3 What a filing emits

A filing carries two signals, and their precisions differ by an order of magnitude.

**The type signal.** A filing is an allegation about  $\theta_j$ . It is informative with probability  $\pi$  and pure noise with probability  $1 - \pi$ , and the voter cannot tell which. Writing  $\tau_I$  for the precision of an informative filing, the effective precision is  $\bar{\tau} = \pi\tau_I$ ; the race emits a public signal  $x_j = \theta_j + e_j$  with  $e_j \sim N(0, 1/\bar{\tau})$  independent across candidates, and the posterior is normal with

$$\hat{\theta}_j = \frac{h_j m_j + \bar{\tau} x_j}{h_j + \bar{\tau}}, \quad \hat{V}(\theta_j) = \frac{1}{h_j + \bar{\tau}}.$$

Genuine information about a candidate and a weapon that happens to persuade enter through the same  $\bar{\tau}$ . They are therefore not separable for the voter, and not separable for the econometrician either. The non-separability that the empirical design has to live with is a property of the environment, not a shortcoming of the instrument.

**The status signal.** The filer is optimizing and targets the highest-ranked opponent, which is the premise Proposition 1 takes from Nakaguma and Souza (2025). So the identity of the target is informative about who is ahead, and its precision  $\tau_{\text{status}}$  is high, because the filer knows the race. A court case also draws local attention, and revealing which candidates are first and second is known to move votes (Anagol and Fujiwara, 2016).

Litigation is thus precise about position and imprecise about quality.

## 4 Consolidation

### 4.1 What $\lambda$ does to the primitives

The modeling content of the whole note sits in one assumption, so it is stated as one rather than smuggled into a proof.

**Assumption 2** (The noisy arena).  $\frac{d\bar{\tau}}{d\lambda} < 0$  and  $\frac{d\tau_{\text{status}}}{d\lambda} > 0$ .

Judicialization crowds the arena with procedural content – propaganda, registration, campaign accounting – that carries little information about whether a candidate would govern well. Raising  $\lambda$  therefore lowers  $\pi$ , the chance that a given filing is about valence at all, and  $\bar{\tau} = \pi\tau_I$  falls with it. The same suits are public, costly, and filed only against opponents worth attacking, so more of them sharpen common knowledge of the rank order. One channel degrades information about *type*; the other improves information about *rank*. That asymmetry is the assumption, and the rest of this section is its consequences.

## 4.2 The type channel

**Lemma 1** (The posterior gap). *Restrict the top two to a common prior precision,  $h_1 = h_2 = h$ ; Proposition 5 relaxes it. Let the priors be independent across candidates. Write  $w = \bar{\tau}/(h + \bar{\tau})$ ,  $\Delta = \hat{\theta}_1 - \hat{\theta}_2$ , and  $\varsigma^2 = 2v_\varepsilon$  for the variance of  $\varepsilon_{i1} - \varepsilon_{i2}$ . Taken ex ante, over the prior on  $(\theta_1, \theta_2)$  as well as over the signal noise,  $\Delta$  is normal with*

$$\mu_\Delta \equiv \mathbb{E}[\Delta] = m_1 - m_2, \quad \sigma_\Delta^2 \equiv \text{Var}(\Delta) = \frac{2\bar{\tau}}{h(h + \bar{\tau})},$$

so  $\mu_\Delta$  carries no  $\bar{\tau}$ , while

$$\frac{d\sigma_\Delta^2}{d\bar{\tau}} = \frac{2}{(h + \bar{\tau})^2} > 0 \quad \text{for every } \bar{\tau} > 0.$$

Suppose  $\mu_\Delta > 0$ , that is, the leader also holds the better reputation. Then

- (a) the probability of a rank reversal is  $\Pr(\Delta < 0) = \Phi(-\mu_\Delta/\sigma_\Delta)$ , strictly increasing in  $\sigma_\Delta$ ;
- (b) the expected two-way margin between the top two is

$$\mathbb{E}[2\Phi(\Delta/\varsigma) - 1] = 2\Phi\left(\frac{\mu_\Delta}{\sqrt{\varsigma^2 + \sigma_\Delta^2}}\right) - 1,$$

strictly decreasing in  $\sigma_\Delta$ .

Under Assumption 2,  $\sigma_\Delta$  is strictly decreasing in  $\lambda$ . So a rise in  $\lambda$  makes rank reversals rarer and the expected two-way margin wider, with no restriction on  $\bar{\tau}$  relative to  $h$  and nothing held fixed.

The frame is ex ante and that is the modeling content, not a technicality: the comparative static is taken across realizations of the campaign, which is the object the estimates correspond to – a cross-section of municipalities that differ in  $\lambda$ , not the single realized draw of one municipality.

*Proof.* With a common  $h$ ,  $\hat{\theta}_j = (1 - w)m_j + wx_j$ , where  $x_j = \theta_j + e_j$  with  $\theta_j \sim N(m_j, 1/h)$  and  $e_j \sim N(0, 1/\bar{\tau})$ , all four draws independent. So  $x_j$  is normal with mean  $m_j$  and variance  $1/h + 1/\bar{\tau}$ , giving  $\mathbb{E}[\hat{\theta}_j] = (1 - w)m_j + w m_j = m_j$ , which carries no  $\bar{\tau}$ , and

$$\text{Var}(\hat{\theta}_j) = w^2 \left( \frac{1}{h} + \frac{1}{\bar{\tau}} \right) = \frac{\bar{\tau}^2}{(h + \bar{\tau})^2} \cdot \frac{h + \bar{\tau}}{h\bar{\tau}} = \frac{\bar{\tau}}{h(h + \bar{\tau})}.$$

Since  $\Delta$  is a linear combination of independent normals it is normal, with  $\mu_\Delta = m_1 - m_2$  and  $\sigma_\Delta^2 = 2\bar{\tau}/[h(h + \bar{\tau})]$ . Differentiating,

$$\frac{d\sigma_\Delta^2}{d\bar{\tau}} = \frac{d}{d\bar{\tau}} \frac{2\bar{\tau}}{h^2 + h\bar{\tau}} = \frac{2(h^2 + h\bar{\tau}) - 2h\bar{\tau}}{(h^2 + h\bar{\tau})^2} = \frac{2h^2}{h^2(h + \bar{\tau})^2} = \frac{2}{(h + \bar{\tau})^2},$$

positive for every  $\bar{\tau} > 0$ . The two limits fix what the expression means:  $\sigma_\Delta^2 \rightarrow 0$  as  $\bar{\tau} \rightarrow 0$ , where the posteriors never leave the reputations, and  $\sigma_\Delta^2 \rightarrow 2/h$  as  $\bar{\tau} \rightarrow \infty$ , where they reach the valences themselves.

(a)  $\Delta \sim N(\mu_\Delta, \sigma_\Delta^2)$ , so  $\Pr(\Delta < 0) = \Phi(-\mu_\Delta/\sigma_\Delta)$  and

$$\frac{\partial \Pr(\Delta < 0)}{\partial \sigma_\Delta} = \varphi\left(\frac{\mu_\Delta}{\sigma_\Delta}\right) \frac{\mu_\Delta}{\sigma_\Delta^2} > 0 \quad \text{whenever } \mu_\Delta > 0.$$

(b) Under the common- $h$  restriction the two ambiguity terms coincide and cancel from the comparison between the top two, so voter  $i$  prefers candidate 1 to candidate 2 exactly when  $\Delta + \varepsilon_{i1} - \varepsilon_{i2} > 0$ . Conditional on  $\Delta$  the two-way margin is therefore  $2\Phi(\Delta/\varsigma) - 1$ . Write  $\Delta = \mu_\Delta + \sigma_\Delta Z$  with  $Z$  standard normal and use  $\mathbb{E}[\Phi(a + bZ)] = \Phi(a/\sqrt{1 + b^2})$ , which follows from  $\Phi(a + bZ) = \Pr(U - bZ \leq a)$  for an independent standard normal  $U$  together with  $U - bZ \sim N(0, 1 + b^2)$ . With  $a = \mu_\Delta/\varsigma$  and  $b = \sigma_\Delta/\varsigma$  the expectation is  $\Phi(\mu_\Delta/\sqrt{\varsigma^2 + \sigma_\Delta^2})$ , strictly decreasing in  $\sigma_\Delta$  when  $\mu_\Delta > 0$ .

Finally  $d\sigma_\Delta/d\lambda = (d\sigma_\Delta/d\bar{\tau})(d\bar{\tau}/d\lambda) < 0$ : the first factor is positive for every  $\bar{\tau} > 0$  and the second is negative by Assumption 2.  $\square$

*Remark* (Why this is not a Jensen argument). Part (b) is not obtained by calling the map from  $\Delta$  to the margin concave. That map is concave on the positive half-line and convex on the negative one, so a mean-preserving contraction of a distribution with support on both sides gets no unambiguous sign from curvature alone, and an argument that took one would be assuming its conclusion. What delivers the sign is the exact Gaussian convolution identity, which needs no support condition. The reading is also sharper than a curvature argument would give: noise does not favor the leader directly. It supplies the only route by which the runner-up could have overtaken, and lowering it closes that route. Part (a) says the same thing without any vote-share mapping at all.

*Remark* (Why the lemma is stated ex ante, and what is lost). Conditioning on the realized valences instead gives  $\mu_\Delta = (1 - w)(m_1 - m_2) + w(\theta_1 - \theta_2)$ , which is a function of  $w$  and therefore of  $\bar{\tau}$ :

$$\frac{d\mu_\Delta}{d\bar{\tau}} = \frac{h}{(h + \bar{\tau})^2} [(\theta_1 - \theta_2) - (m_1 - m_2)].$$

Conditional on a leader whose true valence advantage exceeds its reputational advantage, that derivative is positive, so a fall in  $\bar{\tau}$  lowers  $\mu_\Delta$  and the mean channel runs *against* the variance channel. The variance also changes frame: conditioning on the realized valences leaves only  $e_1, e_2$  random in  $\hat{\theta}_j = (1 - w)m_j + w(\theta_j + e_j)$ , so the conditional-frame variance of  $\Delta$  is

$$\sigma_{\Delta, \text{cond}}^2 \equiv \text{Var}(\Delta \mid \theta_1, \theta_2) = \frac{2\bar{\tau}}{(h + \bar{\tau})^2},$$

not the ex ante  $\sigma_\Delta^2 = 2\bar{\tau}/[h(h + \bar{\tau})]$  of Lemma 1. The numbers below use this conditional-frame variance. The mean channel can also dominate the variance channel: at  $h = 1$ ,  $v_\varepsilon = 1$ ,  $m_1 - m_2 = 0.01$  and  $\theta_1 - \theta_2 = 1$ , moving  $\bar{\tau}$  from 0.5 to 0.4 raises the conditional reversal probability from 0.305 to 0.323 and lowers the conditional expected margin from 0.172 to 0.150 – both against Lemma 1. So the conditional statement has no sign, and the lemma is not merely more convenient stated ex ante: it is only true stated ex ante. What is lost is any claim about a single race whose valences are already drawn; what is kept is the claim about the cross-section, which is what the estimates measure.

### 4.3 The coordination channel

The second leg is a standard result and is cited here rather than re-derived. As  $\tau_{\text{status}}$  rises the rank order approaches common knowledge, and in a coordination equilibrium voters desert candidates ranked third and below, with the desertion running toward the believed leader (Myerson and Weber, 1993; Fey, 1997; Cox, 1997). We import it in reduced form: a fraction  $g(\tau_{\text{status}}) \in [0, 1]$  of the voters attached to candidates ranked third and below deserts, with  $g' > 0$ , and a fraction  $q > 1/2$  of those deserters moves to the believed leader.

### 4.4 The proposition

**Proposition 2** (Consolidation). *Let  $n \in [0, 1]$  be the mass of voters attached to candidates ranked third and below. Under Assumptions 1 and 2, with  $h_1 = h_2 = h$  and  $\mu_\Delta > 0$ , the expected top-two margin is*

$$\mathbb{E}[s_1 - s_2] = (1 - n) \left[ 2\Phi\left(\frac{\mu_\Delta}{\sqrt{\varsigma^2 + \sigma_\Delta^2}}\right) - 1 \right] + n g(\tau_{\text{status}}) (2q - 1),$$

and

$$\frac{d\mathbb{E}[s_1 - s_2]}{d\lambda} > 0.$$

*The probability of a rank reversal falls, so the identity of the leader is more likely to be preserved rather than less, and the size of the field is unchanged.*

*Proof.* The mass  $1 - n$  attached to the top two splits between them, and by Lemma 1(b) its expected contribution to the margin is  $(1 - n)[2\Phi(\mu_\Delta/\sqrt{\varsigma^2 + \sigma_\Delta^2}) - 1]$ . Of the mass  $n$  attached to the rest, a fraction  $g$  deserts,  $q$  of it to candidate 1 and  $1 - q$  to candidate 2, contributing  $n g (q - (1 - q)) = n g (2q - 1)$ . Adding gives the display.

Differentiate. The first term moves through  $\sigma_\Delta$  alone, because by Lemma 1 the ex ante  $\mu_\Delta$  equals  $m_1 - m_2$  and so carries no  $\bar{\tau}$  and no  $\lambda$ ; and

$$\frac{\partial}{\partial \sigma_\Delta} \left[ 2\Phi\left(\frac{\mu_\Delta}{\sqrt{\varsigma^2 + \sigma_\Delta^2}}\right) - 1 \right] \cdot \frac{d\sigma_\Delta}{d\lambda} > 0,$$

because the bracket is strictly decreasing in  $\sigma_\Delta$  by Lemma 1(b) while  $d\sigma_\Delta/d\lambda < 0$  by Assumption 2. The second term has derivative  $n(2q - 1) g'(\tau_{\text{status}}) d\tau_{\text{status}}/d\lambda \geq 0$ , since  $q > 1/2$ ,  $g' > 0$  and  $d\tau_{\text{status}}/d\lambda > 0$  by Assumption 2, and it is strictly positive when  $n > 0$ . Both channels are non-negative and the first is strictly positive, so the sum is. The reversal statement is Lemma 1(a) with  $\sigma_\Delta$  falling in  $\lambda$ . The field is fixed because registration closes before the filings (Section 1).  $\square$

The candidate-supply null is structural rather than incidental: the status signal arrives after the registration deadline, so it cannot act on the candidate set.

**Proposition 3** (Concentration indices are redundant). *Write  $p_1$  and  $p_2$  for the observed shares of the leader and the runner-up, and let the shift be a one-parameter reallocation between them:  $p_1 \mapsto p_1 + \eta$  and  $p_2 \mapsto p_2 - \eta$ , with every other candidate's share unchanged. Then*

- (i) the combined top-two share  $p_1 + p_2$  is invariant in  $\eta$ ;
- (ii)  $\text{HHI}(\eta) = \text{HHI}(0) + 2\eta(p_1 - p_2) + 2\eta^2$ , so  $d\text{HHI}/d\eta = 2(p_1 - p_2 + 2\eta)$ , which changes sign at  $\eta = -(p_1 - p_2)/2$ ; the index is U-shaped in  $\eta$  and monotone only on  $\eta \geq -(p_1 - p_2)/2$ , which is exactly the region where the leader stays ahead;
- (iii) on that region the effective number of candidates  $1/\text{HHI}$  is strictly decreasing in  $\eta$ .

The margin itself is  $(p_1 - p_2) + 2\eta$ . So on the region where the leader stays ahead each index is a deterministic function of the same single parameter, and none of them carries a restriction independent of the margin.

*Proof.* (i)  $(p_1 + \eta) + (p_2 - \eta) = p_1 + p_2$  for every  $\eta$ .

(ii) Expanding,  $(p_1 + \eta)^2 + (p_2 - \eta)^2 = p_1^2 + p_2^2 + 2\eta(p_1 - p_2) + 2\eta^2$ ; the other candidates' shares enter HHI additively and are unchanged. Differentiating gives  $2(p_1 - p_2) + 4\eta$ , which vanishes at  $\eta = -(p_1 - p_2)/2$  and is positive above it. The leader stays ahead when  $p_1 + \eta \geq p_2 - \eta$ , that is when  $\eta \geq -(p_1 - p_2)/2$ : the same threshold.

(iii)  $x \mapsto 1/x$  is strictly decreasing on  $x > 0$ , and HHI is strictly increasing in  $\eta$  there. Finally  $(p_1 + \eta) - (p_2 - \eta) = (p_1 - p_2) + 2\eta$ .  $\square$

This is why concentration cannot identify which mechanism is at work. The point is not that the concentration estimates are imprecise; it is that even a precise one would be the margin restated – and in the case of the combined top-two share it would carry no information at all. Pushing the estimated top-two reallocation through each municipality's own 2024 share vector implies  $\Delta\text{HHI} = +0.018$  and  $\Delta\text{ENP} = -0.055$ , while the separately estimated index regressions give  $+0.023$  (0.013) and  $-0.044$  (0.039). Both implied values sit inside the estimated intervals, so the index regressions add no restriction to the margin result (`docs/model/check_proposition3.py`).

## 5 Exit and the seat split

### 5.1 Exit

**Proposition 4** (Exit). *Once the status signal makes the rank order common knowledge the pivot probability collapses and the choice becomes expressive: voter  $i$  marks the candidate solving*

$$\max_j \left[ \hat{\theta}_j - \frac{\rho/2}{h_j + \bar{\tau}} + \varepsilon_{ij} \right]$$

*if that maximum is at least  $\bar{b}$ , and marks blank otherwise. Write  $c_j = \bar{b} - \hat{\theta}_j + (\rho/2)/(h_j + \bar{\tau})$ . Holding the posterior means fixed,*

$$\frac{d\Pr(\text{blank})}{d\lambda} = -\frac{d\bar{\tau}}{d\lambda} \sum_j \frac{\rho/2}{(h_j + \bar{\tau})^2} \frac{\varphi(c_j/\sqrt{v_\varepsilon})}{\sqrt{v_\varepsilon}} \prod_{k \neq j} \Phi(c_k/\sqrt{v_\varepsilon}) > 0$$

*under Assumption 2, and the effect is exactly zero at  $\rho = 0$ . Under compulsory voting the exit option is a blank or null vote rather than abstention, so turnout is unmoved.*

*Proof.* The voter blanks when every candidate falls short of  $\bar{b}$ , that is when  $\varepsilon_{ij} < c_j$  for all  $j$ . The  $\varepsilon_{ij}$  are independent  $N(0, v_\varepsilon)$ , so  $\Pr(\text{blank}) = \prod_j \Phi(c_j/\sqrt{v_\varepsilon})$ . Holding  $\hat{\theta}_j$  fixed,  $\partial c_j/\partial \bar{\tau} = -(\rho/2)(h_j + \bar{\tau})^{-2} \leq 0$ , so

$$\frac{\partial \Pr(\text{blank})}{\partial \bar{\tau}} = \sum_j \frac{\varphi(c_j/\sqrt{v_\varepsilon})}{\sqrt{v_\varepsilon}} \frac{\partial c_j}{\partial \bar{\tau}} \prod_{k \neq j} \Phi(c_k/\sqrt{v_\varepsilon}) \leq 0,$$

strictly negative when  $\rho > 0$  and identically zero when  $\rho = 0$ . Multiplying by  $d\bar{\tau}/d\lambda < 0$  gives the display and its sign. Conditioning on the posterior means is what isolates the ambiguity channel; their own movement in  $\bar{\tau}$  is the type channel, which Proposition 2 already treats, and it leaves  $\mathbb{E}[\hat{\theta}_j] = m_j$  for every  $\bar{\tau}$ .  $\square$

The mechanism is the posterior variance.  $\hat{V}(\theta_j) = 1/(h_j + \bar{\tau})$  is decreasing in  $\bar{\tau}$ , so under Assumption 2 a rise in  $\lambda$  raises the posterior variance of *every* candidate's valence, and an ambiguity-averse voter discounts every candidate by more. The maximum falls, more voters find no name worth  $\bar{b}$ , and the blank rate rises. The blank response and the consolidation are then the same single parameter movement seen from two sides –  $\bar{\tau}$  falling widens the margin through Lemma 1 and empties the ballot through Proposition 4. Because the effect vanishes at  $\rho = 0$ , the blank rise is the testable content of ambiguity aversion rather than a restatement of it (Ghirardato and Katz, 2006); and compulsory voting is what routes the response into the mark rather than into attendance, which is the margin the abstention literature works on (Feddersen and Pesendorfer, 1996).

## 5.2 The seat split

The seat dimension enters through one primitive, the prior precision on the leader's valence. In a contested seat an incumbent is running, so  $h_1$  is high: voters have already learned  $\theta_1$  and have already sorted on it. In an open seat there is no incumbent, so  $h_1$  is low and there is nothing to have sorted on.

**Lemma 2** (Selection among the runner-up's supporters). *Under Assumption 1, voter  $i$  starts out supporting candidate 2 exactly when*

$$D_i \equiv (\varepsilon_{i1} + \zeta_{i1}) - (\varepsilon_{i2} + \zeta_{i2}) < -\delta, \quad \delta \equiv \left(m_1 - \frac{\rho/2}{h_1}\right) - \left(m_2 - \frac{\rho/2}{h_2}\right),$$

and  $\delta \geq 0$  is equivalent to  $s_1^\circ \geq s_2^\circ$ . Write  $\sigma_D^2 = 2v_\varepsilon + \kappa_\zeta/h_1 + \kappa_\zeta/h_2$ ,  $z = \delta/\sigma_D$  and  $r = \sqrt{v_\varepsilon}/\sigma_D = \text{corr}(\varepsilon_{i1}, D_i)$ . Then

(a)  $\mathbb{E}[\varepsilon_{i1} \mid D_i < -\delta] = -\frac{v_\varepsilon}{\sigma_D} \frac{\varphi(z)}{\Phi(-z)}$ , which is strictly increasing in  $\sigma_D$  and hence strictly decreasing in  $h_1$ ;

(b) for every threshold  $t$ ,  $\Pr(\varepsilon_{i1} \geq t \mid D_i < -\delta)$  is strictly decreasing in  $h_1$ .

*Proof.*  $D_i$  is symmetric about zero, so  $s_1^\circ = \Pr(D_i \geq -\delta) \geq 1/2$  exactly when  $\delta \geq 0$ ; that is the labeling of Section 1 and not an extra hypothesis.

(a)  $(\varepsilon_{i1}, D_i)$  is bivariate normal with  $\text{Cov}(\varepsilon_{i1}, D_i) = v_\varepsilon$ , so  $\mathbb{E}[\varepsilon_{i1} \mid D_i] = (v_\varepsilon/\sigma_D^2) D_i$  and

$$\mathbb{E}[\varepsilon_{i1} \mid D_i < -\delta] = \frac{v_\varepsilon}{\sigma_D^2} \mathbb{E}[D_i \mid D_i < -\delta] = \frac{v_\varepsilon}{\sigma_D^2} \left(-\sigma_D \frac{\varphi(z)}{\Phi(-z)}\right) = -\frac{v_\varepsilon}{\sigma_D} M(z),$$

where  $M(z) = \varphi(z)/\Phi(-z)$  is the standard normal hazard rate, which is positive and strictly increasing. Raising  $\sigma_D$  lowers  $v_\varepsilon/\sigma_D$  and, because  $\delta \geq 0$ , lowers  $z$  and so does not raise  $M(z)$ ; the product falls and its negative rises. And  $\sigma_D$  is strictly decreasing in  $h_1$ .

(b) Standardize. With  $a = t/\sqrt{v_\varepsilon}$  the quantity is  $\Pr(Z_1 \geq a \mid Z_2 \leq -z)$  for a standard bivariate normal pair with correlation  $r > 0$ , and it is decreasing in each of  $z$  and  $r$ . In  $z$ :

$$\Pr(Z_1 \geq a \mid Z_2 \leq -z) = \mathbb{E} \left[ \bar{\Phi} \left( \frac{a - rZ_2}{\sqrt{1 - r^2}} \right) \mid Z_2 \leq -z \right],$$

whose integrand is increasing in  $Z_2$  because  $r > 0$ , while the conditional law of  $Z_2$  given  $Z_2 \leq -z$  falls in the first-order stochastic order as  $z$  rises. In  $r$ : by Plackett's identity,  $\partial \Pr(Z_1 \leq a_1, Z_2 \leq a_2)/\partial r$  equals the bivariate normal density at  $(a_1, a_2)$  and is therefore positive; applying it to  $\Pr(-Z_1 \leq -a, Z_2 \leq -z)$ , whose pair has correlation  $-r$ , makes the numerator decreasing in  $r$ , while the denominator  $\Phi(-z)$  does not depend on  $r$ . Raising  $h_1$  lowers  $\sigma_D$ , which raises both  $z$  and  $r$ , so it lowers the conditional probability.  $\square$

**Proposition 5** (The seat split). *Take the runner-up's departing supporters, the mass that Proposition 2 moves, and assume that mass is comparable across seat types – a restriction on magnitudes that Proposition 2, derived at  $h_1 = h_2 = h$ , does not deliver for seats that differ in  $h_1$ . Each of those supporters either marks candidate 1 or marks blank, by the rule of Proposition 4. At a common threshold – in particular at  $\rho = 0$  – the fraction that marks candidate 1 is strictly lower where the seat is contested. Under that assumption, then, what differs between seat types is where the departing mass goes: to the winner in an open seat, with the valid vote holding, and split between the winner and the blank ballot in a contested seat.*

*Proof.* First, who departs. Proposition 2's display allocates the top-two mass by a probit rule on  $\Delta + \varepsilon_{i1} - \varepsilon_{i2}$  and never names a departing set, so the set is named here: the voters who leave the runner-up are candidate 2's camp  $\{i : D_i < -\delta\}$  of Lemma 2, or a uniform random subsample of it, which is what makes the conditional law of  $\varepsilon_{i1}$  among them the law that lemma computes. That is an assumption about who departs, not a consequence of the display. So is its size: Proposition 2 is derived at  $h_1 = h_2 = h$ , while this proposition turns on  $h_1$  differing by seat, so treating that departing mass as comparable across seat types is a further assumption about magnitudes, not something Proposition 2 delivers.

A departing supporter of candidate 2 marks candidate 1 when  $\hat{\theta}_1 - (\rho/2)/(h_1 + \bar{\tau}) + \varepsilon_{i1} \geq \bar{b}$ , that is when  $\varepsilon_{i1} \geq t$  with  $t = \bar{b} - \hat{\theta}_1 + (\rho/2)/(h_1 + \bar{\tau})$ , and blanks otherwise. The fraction marking candidate 1 is therefore  $\Pr(\varepsilon_{i1} \geq t \mid D_i < -\delta)$ , which at any fixed  $t$  is strictly decreasing in  $h_1$  by Lemma 2(b). A contested seat is a high- $h_1$  seat and an open seat a low- $h_1$  seat, which gives the ordering.

For that to be the whole of the effect the threshold has to be the same in the two seats, and at  $\rho = 0$  it is not automatically:  $t$  reduces to  $\bar{b} - \hat{\theta}_1$ , which still carries  $h_1$  through  $\hat{\theta}_1 = (h_1 m_1 + \bar{\tau} x_1)/(h_1 + \bar{\tau})$ , running from  $x_1$  toward  $m_1$  as  $h_1$  grows. Hold  $\hat{\theta}_1$  at a common realized value across the two seat types: this is the same device Proposition 4's proof uses, holding a posterior mean fixed to isolate one channel from  $\hat{\theta}_1$ 's own movement. At that common value the  $\rho = 0$  threshold  $\bar{b} - \hat{\theta}_1$  is identical in both seats, and the comparison is exactly Lemma 2(b): two races that share a realized leader standing and differ only in whether an incumbent is running, so  $h_1$  alone carries the seat difference through  $\sigma_D$ . Taking the ex ante frame of Lemma 1 instead – averaging  $\Pr(\varepsilon_{i1} \geq t \mid D_i < -\delta)$  over  $\hat{\theta}_1$ 's own law rather than fixing it – would not deliver this:  $\text{Var}(\hat{\theta}_1) = \bar{\tau}/[h_1(h_1 + \bar{\tau})]$  is decreasing in  $h_1$ , so the open seat's threshold is the more

dispersed one, and  $\mathbb{E}[\Pr(\varepsilon_{i1} \geq t \mid D_i < -\delta)]$  need not equal  $\Pr(\varepsilon_{i1} \geq \mathbb{E}[t] \mid D_i < -\delta)$  inside that nonlinear probability – the same substitution the remark “Why this is not a Jensen argument” above declines to make for Lemma 1(b).  $\square$

The split comes from selection on  $\varepsilon$  alone, and it is worth being explicit about what the ambiguity term of Proposition 4 does here, because it does not generate the split and does not even help. That term raises the blank rate in *both* kinds of seat, so on its own it predicts no contrast. Worse, its level differs in the unhelpful direction:  $\hat{V}(\theta_1) = 1/(h_1 + \bar{\tau})$  is largest exactly where  $h_1$  is smallest, so the ambiguity penalty on the leader is heaviest in the open seat, which is where the model predicts the least blanking. Formally,  $t$  is higher in the open seat when  $\rho > 0$ . The prediction therefore holds outright at  $\rho = 0$  and, for  $\rho > 0$ , requires the selection effect of Lemma 2(b) to dominate that differential penalty. Saying so is the honest statement of it.

The reading is the one the mechanism suggests. Where the leader is a sitting mayor, still supporting the challenger is a revealed rejection of someone the whole electorate has had four years to judge, so those voters are selected on genuinely low taste for the incumbent, and a large share of them will not mark that name under any amount of coordination pressure. Where the leader is fresh, starting out with the runner-up says much less, because the pre-campaign split was driven by impressions that the campaign then washes out; those voters’ tastes are close to unselected, and most of them follow the coordination.

## 6 What the model does not do

The model produces signs and orderings. It is not calibrated and not structurally estimated. It does not separate a genuine allegation from a persuasive weapon, and Section 3 is the reason that is a property of the environment rather than a gap to be closed. It does not endogenize the filer’s decision: the attack direction is taken from Nakaguma and Souza (2025) as a maintained assumption, because endogenizing it would require exactly the selection the instrument exists to sidestep. It says nothing about judges, and nothing about welfare. Runoff municipalities, where a second round changes the coordination problem, are outside it.

Six of its steps are assumed rather than derived, and it is better to name them than to let them read as results. The attack direction imported from Nakaguma and Souza (2025), noted just above, is the first: Section 3’s status channel and Proposition 1’s foil both rest on it. Assumption 2 is the whole modeling content of Section 4: nothing here shows that  $\lambda$  must push  $\bar{\tau}$  down and  $\tau_{\text{status}}$  up, only that everything else follows if it does. The second half of Assumption 1 – that pre-campaign impressions decide the initial camps but are washed out of the ballot-stage payoff – is what makes the seat split run in the predicted direction; without it the selected object in Lemma 2 would be the composite  $\varepsilon_{i1} + \zeta_{i1}$  rather than  $\varepsilon_{i1}$ , and the selection would reverse, because a fresh leader’s more dispersed impressions would themselves make the runner-up’s camp more extreme. The identity of the departing set is assumed too, in Proposition 5’s proof: Proposition 2 moves mass between the top two without naming who moves, so taking the movers to be candidate 2’s camp, or a uniform subsample of it, is what lets Lemma 2 apply to them. So is the size of that mass: Proposition 2 is derived at  $h_1 = h_2 = h$ , while Proposition 5 turns on  $h_1$  differing by seat, so its claim that the runner-up loses a comparable mass in either kind of seat is a further assumption, not a consequence of Proposition 2. And the coordination leg of Proposition 2 is imported from the literature rather than derived (Myerson and Weber, 1993; Fey, 1997; Cox, 1997).

One further limit is a matter of what the statements are about rather than of what is assumed. Lemma 1 and everything downstream of it are *ex ante*, taken over the prior on valences as well as over the signal noise, which matches a cross-section of municipalities differing in  $\lambda$  but says nothing about a race whose valences are already drawn. Conditional on those, the remark *Why the lemma is stated ex ante* shows the two comparative statics have no sign at all. Two results are exceptions, and they are exceptions in different ways. Proposition 4’s blank-vote result holds the posterior means  $\hat{\theta}_j$  fixed and moves only  $\bar{\tau}$  through the ambiguity term, so it is a within-race comparative static for a race whose valences are already drawn. Proposition 5’s seat comparison holds  $\hat{\theta}_1$  at a common realized value and compares two races that differ in  $h_1$ , so it is a between-race statement conditional on that realization. Neither averages over the prior on valences.

Two conditions bound the mechanism. Coordination needs a small number of focal contenders, so a genuinely three-way race weakens it. And the reasoning in Section 4 applies to a single-winner plurality contest; an open-list proportional council race offers no focal rank for the status signal to reveal, which is the model’s clearest out-of-sample implication.

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